PROBLEM STATEMENT 04 · CONTRIBUTOR TRACK

# Build a Feature the Open-Source Gateway Is Missing

AI gateways are becoming essential infrastructure for teams building production AI applications. Developers need gateways that support their preferred model providers, integrate with existing databases and development tools, and provide flexible routing, caching, and observability.

Bifrost offers extension points that developers can use to introduce new capabilities into the open-source gateway.

## GOAL

**Extend Bifrost with a practical feature that solves a real operational or developer problem: a plugin, integration, UI, routing or caching strategy, or developer tool that could potentially be contributed upstream.**

## MANDATORY REQUIREMENTS

- **Real operator problem:** Identify a meaningful limitation or missing capability and solve it end to end.
- **Public extension points:** Build using supported interfaces such as plugins, logging APIs, routing rules, or the MCP gateway.
- **Reproducible integration:** Demonstrate the feature working with a standard Bifrost installation and provide clear setup instructions.
- **Contribution quality:** Include appropriate tests, documentation, and a repository prepared for a potential upstream pull request.

## EXPECTED OUTCOME

A working, documented, and tested Bifrost extension that solves a meaningful problem for gateway operators or developers and is suitable for consideration as an open-source contribution.

## OPTIONAL IMPLEMENTATION GUIDANCE

A new model-provider integration, a Postgres/pgvector caching backend, a gateway migration utility, a prompt optimization plugin, or a VS Code extension displaying usage and remaining budgets. These are examples, not mandatory features; check existing functionality before selecting a contribution.